

Modern Macroeconometrics

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Outline

Many of the time series and econometric tools that have long been workhorses of empirical macroeconomic analysis at central banks were largely developed for an environment with comparatively few series and stable economic relationships. Two recent developments mean that these tools, while still useful, are no longer sufficient on their own: Big Data and machine learning. By Big Data we refer to the large and rapidly growing number of macroeconomic and financial time series available to researchers and policymakers. Where it was once common to build regressions or vector autoregressions (VAR) using only a handful of variables, it is now routine to specify models with dozens or even hundreds of variables. By machine learning we refer to the idea that, with appropriate models and software, computers can uncover relationships between variables even when those relationships are highly complex (e.g., due to a rapidly changing economic environment, financial crises, or the Covid-19 pandemic).

In macroeconomics, especially in the presence of Big Data, there is a real risk that any single class of model will not be rich enough empirically. This has motivated a growing interest in Bayesian and machine learning methods, often based on either parametric (possibly time-varying) models with adaptive shrinkage/regularisation or fully nonparametric models, which allow for nonlinear relations between dependent and explanatory variables without imposing a specific functional form, letting the data decide instead. In addition, in particular post-Covid, modelling techniques for handling heteroskedasticity, outliers, and risk are crucial in any applied macroeconomic toolkit nowadays.

This 3-day course provides a hands-on introduction to modern Bayesian macroeconometrics, with a particular emphasis on tools relevant to central bank research and policy analysis. The course comprises 11 hours of lectures delivered by Dr. Hauzenberger, combining relevant theoretical concepts with applications to real-world macroeconomic and financial datasets. All course materials will be made available via [GitHub](#), including associated **R** code for the empirical illustrations.

Schedule

- **Day 1 – Tuesday, 7 July 2026:** morning session, 10:00–12:15 (including a 15-minute break); afternoon session, 14:00–16:15 (including a 15-minute break)
- **Day 2 – Wednesday, 8 July 2026:** morning session, 10:00–12:15 (including a 15-minute break); afternoon session, 14:00–17:15 (including a 15-minute break)
- **Day 3 – Thursday, 9 July 2026:** morning session, 10:00–12:15 (including a 15-minute break)

Topics to cover

1. Recent themes in macroeconometrics (Day 1, morning session)

A high-level overview and motivation for the course programme. Why Bayesian machine learning methods are attractive for research at central banks, discussing Big Data, machine learning, nonlinearities, and risk monitoring (the macroeconomy at risk).

2. Bayesian linear regression model with Big Data (Day 1, afternoon session)

A primer and overview of Bayesian econometrics using the linear regression model as the leading example. The block briefly covers the basics of Bayesian theory and computation and explains why Bayesian shrinkage and regularisation are attractive for macroeconomic and financial research (small samples, many candidate predictors, formal uncertainty quantification).

Readings: Standard introductory textbooks on Bayesian econometrics that cover the linear regression model include [Koop \(2003\)](#), [Lancaster \(2004\)](#) and [Greenberg \(2012\)](#). [Chan *et al.* \(2019\)](#) is a companion book of solved exercises providing the underlying derivations and proofs. Classic references on adaptive shrinkage and variable selection include [George and McCulloch \(1993\)](#) for the Stochastic Search Variable Selection (SSVS) prior, [Brown and Griffin \(2010\)](#) for the Normal-Gamma prior, [Park and Casella \(2008\)](#) for the Bayesian Lasso, and [Carvalho *et al.* \(2009\)](#), [Carvalho *et al.* \(2010\)](#) as well as [Makalic and Schmidt \(2015\)](#) for the Horseshoe prior. Recent macroeconomic applications include [Huber and Feldkircher \(2019\)](#), [Follett and Yu \(2019\)](#), [Hauzenberger *et al.* \(2024\)](#), and [Gruber and Kastner \(2025\)](#).

3. Bayesian state space modelling techniques (Day 2, morning session)

Many of the workhorse models in macroeconometrics (e.g., unobserved-components models, time-varying parameter models, dynamic factor models, stochastic volatility models, DSGE models, mixed-frequency models, etc.) can be written as Gaussian linear state space models. The focus is on time-varying parameter (TVP) regressions equipped with shrinkage priors that let the data decide which coefficients actually move.

Readings: [Kim and Nelson \(1999\)](#) is a classic reference. [Chan and Strachan \(2023\)](#) provides a useful recent survey of Bayesian state space models, and [Chan *et al.* \(2023\)](#) develops efficient computational methods with a particular focus on mixed-frequency and missing data. Key references for shrinkage in TVP regressions are [Frühwirth-Schnatter and Wagner \(2010\)](#), [Dangl and Halling \(2012\)](#), [Bitto and Frühwirth-Schnatter \(2019\)](#), [Kowal *et al.* \(2019\)](#), [Chan *et al.* \(2020\)](#), [Huber *et al.* \(2021\)](#), [Fischer *et al.* \(2023\)](#), and [Hauzenberger *et al.* \(2022\)](#).

4. Modelling heteroskedasticity and outliers (Day 2, afternoon session)

Modelling techniques for handling time-varying volatility, extreme observations, and asymmetric tail risks. The block covers stochastic volatility (SV) as a non-linear, non-Gaussian state space model; outlier-augmented SV models, which have proved especially useful for post-Covid macro data.

Readings: [Clark and Mertens \(2023\)](#) and [Chan \(2024\)](#) provide a great overview. Other key references include [Clark \(2011\)](#), [Clark and Ravazzolo \(2015\)](#), [Kastner \(2019\)](#), [Chan \(2020\)](#), [Lenza and Primiceri \(2022\)](#), and [Carriero *et al.* \(2024\)](#).

5. The macroeconomy at risk (Day 2, afternoon session)

Bayesian quantile regression for monitoring downside risks (growth) or upside (inflation) risks.

Readings: Key references include [Adrian *et al.* \(2019\)](#), [Adams *et al.* \(2021\)](#), [Pfarrhofer \(2022\)](#), [Lopez-Salido and Loria \(2024\)](#), [Korobilis and Schröder \(2025\)](#), and [Loria *et al.* \(2025\)](#).

6. Bayesian nonparametric methods (Day 3, morning session)

Flexible Bayesian nonparametric methods allow the data to decide on the form of nonlinearities, including Bayesian additive regression trees (BART), Gaussian process (GP) regression, and Bayesian neural networks. These methods can also be embedded in vector autoregressions (VARs), which can be viewed as systems of Bayesian regressions.

Readings: [Williams and Rasmussen \(2006\)](#), [Goodfellow *et al.* \(2016\)](#), [Chipman *et al.* \(2010\)](#), and [Hill *et al.* \(2020\)](#) provide an overview. Key references for applications in macroeconomics include [Huber *et al.* \(2023\)](#), [Clark *et al.* \(2023\)](#), [Marcellino and Pfarrhofer \(2024\)](#), [Hauzenberger *et al.* \(2025b\)](#), and [Hauzenberger *et al.* \(2025a\)](#).

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